KAIZEN MATHS

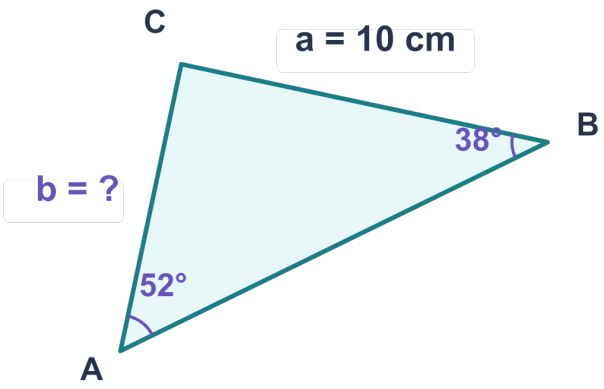
Lesson starter

Sine Rule, Cosine Rule and Triangle Area

Sine Rule & Area

Name _____

SA1



In triangle ABC, A = 52°, B = 38°, and a = 10 cm. Find b to 2 d.p.

ALG-2B2B-L1 · SA

Show your working

KAIZEN MATHS

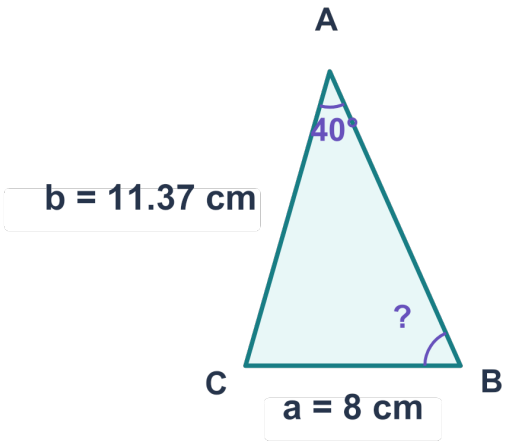
Lesson starter

Sine Rule, Cosine Rule and Triangle Area

Sine Rule & Area

Name _____

SB1



In triangle ABC, A = 40°, a = 8 cm, and b = 11.37 cm. Find all possible values of B to 1 d.p.

ALG-2B2B-L1 · SB

Show your working

KAIZEN MATHS

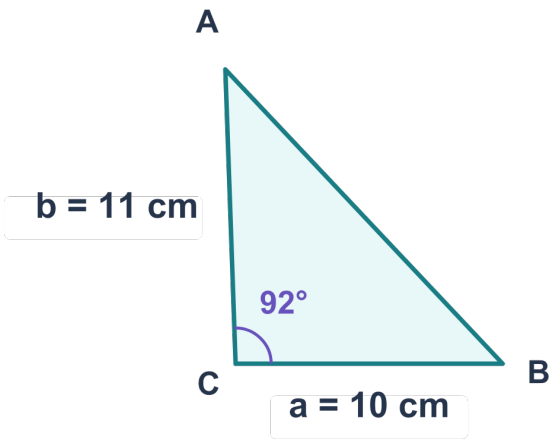
Lesson starter

Sine Rule, Cosine Rule and Triangle Area

Sine Rule & Area

Name _____

SC1



In triangle ABC, a = 10 cm, b = 11 cm, and included angle C = 92°. Find the area to 2 d.p.

ALG-2B2B-L1 · SC

Show your working

KAIZEN MATHS

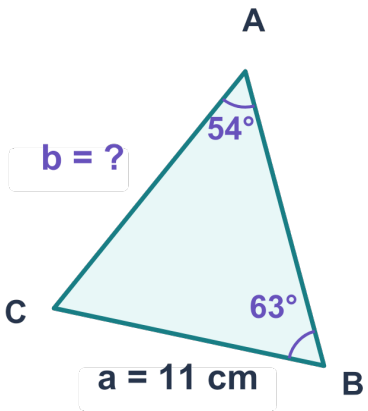
Lesson starter

Sine Rule, Cosine Rule and Triangle Area

Sine Rule & Area

Name _____

SD1



In triangle ABC, A = 54°, B = 63°, and a = 11 cm. Find b to 2 d.p.

ALG-2B2B-L1 · SD

Show your working

KAIZEN MATHS

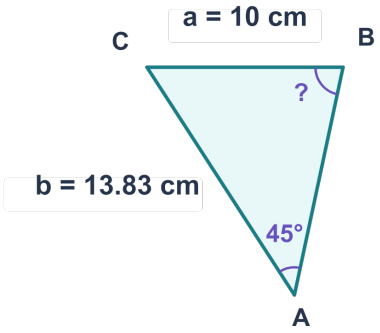
Exit ticket

Sine Rule, Cosine Rule and Triangle Area

Sine Rule & Area

Name _____

EA1



In triangle ABC, A = 45°, a = 10 cm, and b = 13.83 cm. Find all possible values of B to 1 d.p.

Show your working

KAIZEN MATHS

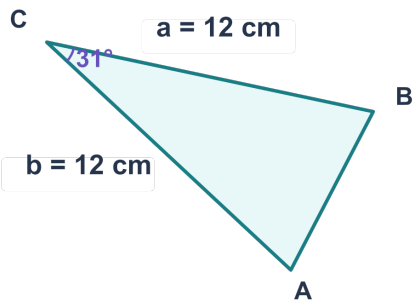
Exit ticket

Sine Rule, Cosine Rule and Triangle Area

Sine Rule & Area

Name _____

EB1



In triangle ABC, a = 12 cm, b = 12 cm, and included angle C = 31°. Find the area to 2 d.p.

Show your working

KAIZEN MATHS

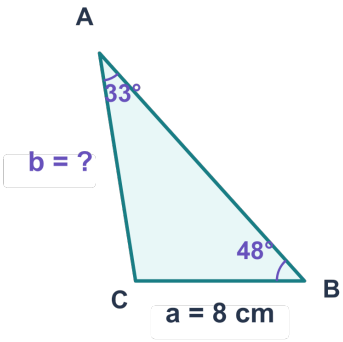
Exit ticket

Sine Rule, Cosine Rule and Triangle Area

Sine Rule & Area

Name _____

EC1



In triangle ABC, A = 33°, B = 48°, and a = 8 cm. Find b to 2 d.p.

Show your working

KAIZEN MATHS

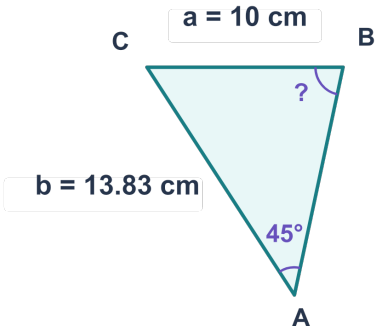
Exit ticket

Sine Rule, Cosine Rule and Triangle Area

Sine Rule & Area

Name _____

EA1



In triangle ABC, A = 45°, a = 10 cm, and b = 13.83 cm. Find all possible values of B to 1 d.p.

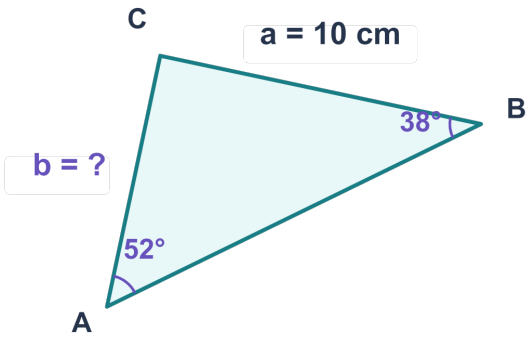
Show your working

Teacher answers & working

Sine Rule, Cosine Rule and Triangle Area · Sine Rule & Area

ALG-2B2B-L1 · Student pages 1-3. Print at 100%, single-sided. Cut on dashed lines.

SA1



In triangle ABC, A = 52°, B = 38°, and a = 10 cm. Find b to 2 d.p.

Answer:

b = 7.81 cm

Working:

Sine Rule: a / sin A = b / sin B

b = (sin B / sin A) × a

b = (sin 38° / sin 52°) × 10

sin 38° ≈ 0.6157, sin 52° ≈ 0.7880

b ≈ 7.8129

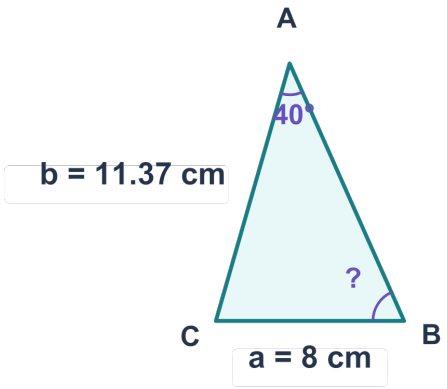
b = 7.81 cm

Teacher answers & working

Sine Rule, Cosine Rule and Triangle Area · Sine Rule & Area

ALG-2B2B-L1 · Student pages 1-3. Print at 100%, single-sided. Cut on dashed lines.

SB1



In triangle ABC, A = 40°, a = 8 cm, and b = 11.37 cm. Find all possible values of B to 1 d.p.

Answer:

$B = 66.0°$ or $114.0°$

Working:

Sine Rule: sin B = (b × sin A) / a

$\sin B = (11.37 \times \sin 40°) / 8 \approx 0.9136$

$B_1 = \sin^{-1}(0.9136) \approx 66.0°$

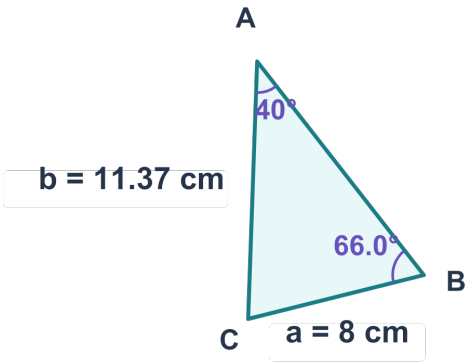
$B_2 = 180° - 66.0023° \approx 114.0°$

$A + B_1 \approx 106.00°$; $A + B_2 \approx 154.00°$

Both sums are below 180°, so there are two possible triangles.

$B = 66.0°$ or $114.0°$

Triangle 1



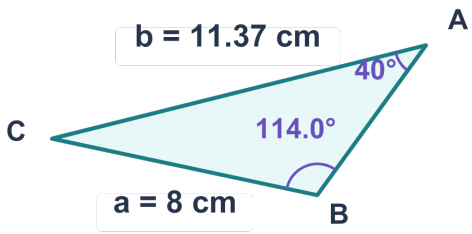
Teacher answers & working

Sine Rule, Cosine Rule and Triangle Area · Sine Rule & Area

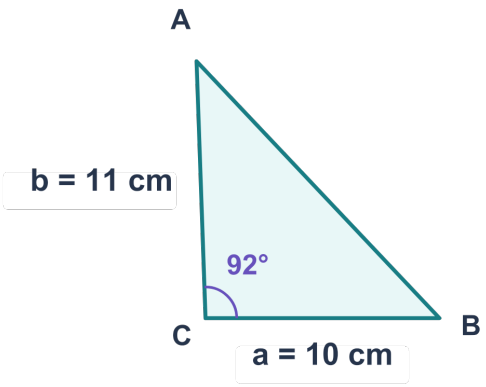
ALG-2B2B-L1 · Student pages 1-3. Print at 100%, single-sided. Cut on dashed lines.

SB1 (continued)

Triangle 2



SC1



In triangle ABC, a = 10 cm, b = 11 cm, and included angle C = 92°. Find the area to 2 d.p.

Answer:

Area = 54.97 cm²

Working:

Area = 1/2 ab sin C

Area = 1/2 × 10 × 11 × sin 92°

sin 92° ≈ 0.9994

Area ≈ 54.9665

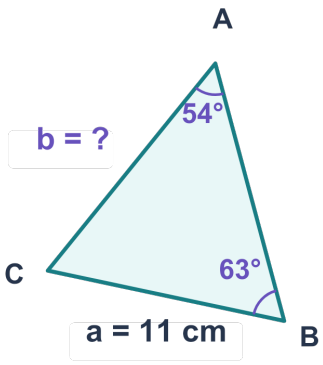
Area = 54.97 cm²

Teacher answers & working

Sine Rule, Cosine Rule and Triangle Area · Sine Rule & Area

ALG-2B2B-L1 · Student pages 1-3. Print at 100%, single-sided. Cut on dashed lines.

SD1



In triangle ABC, A = 54°, B = 63°, and a = 11 cm. Find b to 2 d.p.

Answer:

$b = 12.11$ cm

Working:

Sine Rule: $a / \sin A = b / \sin B$

$b = (\sin B / \sin A) \times a$

$b = (\sin 63° / \sin 54°) \times 11$

$\sin 63° \approx 0.8910$, $\sin 54° \approx 0.8090$

$b \approx 12.1148$

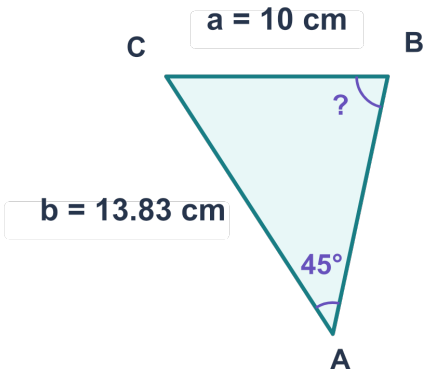
$b = 12.11$ cm

Teacher answers & working

Sine Rule, Cosine Rule and Triangle Area · Sine Rule & Area

ALG-2B2B-L1 · Student pages 1-3. Print at 100%, single-sided. Cut on dashed lines.

EA1



In triangle ABC, A = 45°, a = 10 cm, and b = 13.83 cm. Find all possible values of B to 1 d.p.

Answer:

B = 77.9° or 102.1°

Working:

Sine Rule: sin B = (b × sin A) / a

$\sin B = (13.83 \times \sin 45°) / 10 \approx 0.9779$

$B_1 = \sin^{-1}(0.9779) \approx 77.9°$

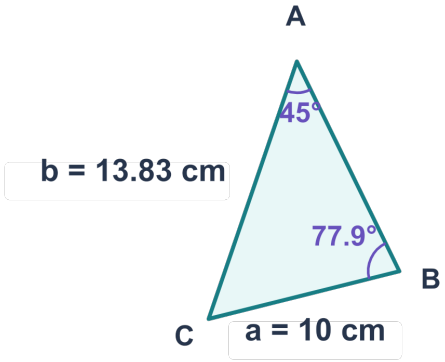
$B_2 = 180° - 77.9398° \approx 102.1°$

$A + B_1 \approx 122.94°$; $A + B_2 \approx 147.06°$

Both sums are below 180°, so there are two possible triangles.

B = 77.9° or 102.1°

Triangle 1



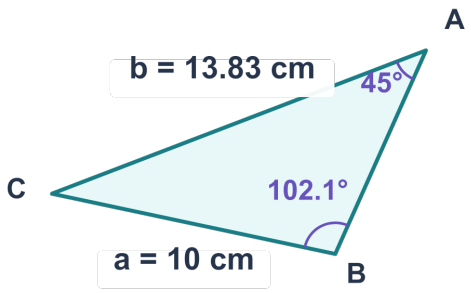
Teacher answers & working

Sine Rule, Cosine Rule and Triangle Area · Sine Rule & Area

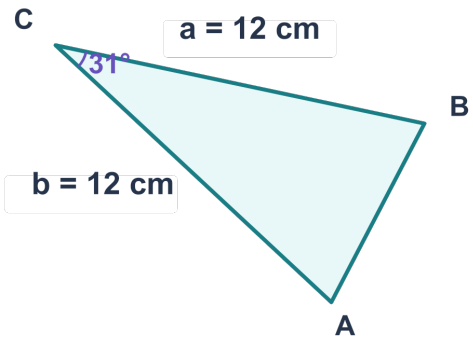
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EA1 (continued)

Triangle 2



EB1



In triangle ABC, a = 12 cm, b = 12 cm, and included angle C = 31°. Find the area to 2 d.p.

Answer:

Area = 37.08 cm²

Working:

Area = 1/2 ab sin C

Area = 1/2 × 12 × 12 × sin 31°

sin 31° ≈ 0.5150

Area ≈ 37.0827

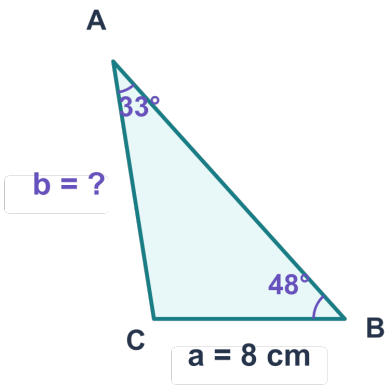
Area = 37.08 cm²

Teacher answers & working

Sine Rule, Cosine Rule and Triangle Area · Sine Rule & Area

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EC1



In triangle ABC, A = 33°, B = 48°, and a = 8 cm. Find b to 2 d.p.

Answer:

b = 10.92 cm

Working:

Sine Rule: a / sin A = b / sin B

b = (sin B / sin A) × a

b = (sin 48° / sin 33°) × 8

sin 48° ≈ 0.7431, sin 33° ≈ 0.5446

b ≈ 10.9158

b = 10.92 cm